

JAVA PROGRAMMING LAB

Experiment No.: 1

Date: 30/07/2026

Aim:

WAP to display a message. "Let us start Java programming."

Program:

```
1 package Program;
2
3 class Programming {
4     void Program() {
5         System.out.println("Let's start Java programming.");
6     }
7 }
8
9 class Main {
10     public static void main(String[] args) {
11         Programming p = new Programming();
12         p.Program();
13     }
14 }
15
16
17
18
```

Output:

```
Let us start Java programming.
```

JAVA PROGRAMMING LAB

Experiment No.: 2

Date: 30/07/2026

Aim:

WAP to find the sum of the two numbers.

Program:

```
1 package Program;
2
3 class Sum {
4
5     void add(int x, int y) {
6
7         System.out.println("sum of two numbers: " + (x + y));
8     }
9
10 }
11
12 class Add {
13
14     public static void main(String[] args) {
15
16         Sum s = new Sum();
17         s.add(10, 15);
18     }
19 }
```

Output:

```
sum of two numbers: 25
```

JAVA PROGRAMMING LAB

Experiment No.: 3

Date: 30/07/2026

Aim:

WAP to understand the effect of print() and println() methods and the backslash codes.

Program:

```
1 package Program;
2
3 class PrintDemo {
4     public static void main(String[] args) {
5
6         System.out.print("Hello ");
7         System.out.print("Java");
8         System.out.print(" Programming");
9
10        System.out.println();
11
12        System.out.println("Hello");
13        System.out.println("Java");
14        System.out.println("Programming");
15
16        System.out.println("Hello\nJava");
17        System.out.println("Hello\tJava");
18        System.out.println("\"Java Programming\"");
19        System.out.println("C:\\Java\\Programs");
20        System.out.println("Hello\bJava");
21        System.out.println("Hello\rJava");
22    }
23 }
```

Output:

```
Hello Java Programming
Hello
Java
Programming
Hello
Java
Hello   Java
"Java Programming"
C:\Java\Programs
HellJava
```


JAVA PROGRAMMING LAB

Experiment No.: 4

Date: 30/07/2026

Aim:

WAP to observe the effects of various bitwise operators.

Program:

```
1 package Program;
2
3 class BitwiseOperator {
4     void operator(String[] args) {
5
6         int a = 5;
7         int b = 3;
8
9         System.out.println("a = " + a);
10        System.out.println("b = " + b);
11
12        System.out.println("a & b = " + (a & b));
13        System.out.println("a | b = " + (a | b));
14        System.out.println("a ^ b = " + (a ^ b));
15        System.out.println("~a = " + (~a));
16        System.out.println("a << 1 = " + (a << 1));
17        System.out.println("a >> 1 = " + (a >> 1));
18    }
19 }
20
21 class Bit {
22     public static void main(String[] args) {
23
24         BitwiseOperator b = new BitwiseOperator();
25         b.operator(args);
26     }
27 }
```

Output:

```
a = 5
b = 3
a & b = 1
a | b = 7
a ^ b = 6
~a = -6
a << 1 = 10
```

$$a \gg 1 = 2$$

JAVA PROGRAMMING LAB

Experiment No.: 5

Date: 30/07/2026

Aim:

WAP to implement the arithmetic operators.

Program:

```
1 package Program;
2
3 class ArithmeticOperators {
4     void operators() {
5
6         int x = 10, y = 5;
7
8         System.out.println("x = " + x);
9         System.out.println("y = " + y);
10
11         System.out.println("x + y = " + (x + y));
12         System.out.println("x - y = " + (x - y));
13         System.out.println("x * y = " + (x * y));
14         System.out.println("x / y = " + (x / y));
15         System.out.println("x % y = " + (x % y));
16
17     }
18 }
19
20 class Arithmetic {
21     public static void main(String[] args) {
22
23         ArithmeticOperators a = new ArithmeticOperators();
24         a.operators();
25     }
26 }
```

Output:

```
x = 10
y = 5
x + y = 15
x - y = 5
x * y = 50
x / y = 2
x % y = 0
```

JAVA PROGRAMMING LAB

Experiment No.: 6

Date: 30/07/2026

Aim:

WAP to implement the relational operators.

Program:

```
1 package Program;
2
3 class RelationalOperators {
4     void operators() {
5         int x = 10, y = 5;
6
7         System.out.println("x = " + x);
8         System.out.println("y = " + y);
9
10        System.out.println("x > y = " + (x > y));
11        System.out.println("x < y = " + (x < y));
12        System.out.println("x >= y = " + (x >= y));
13        System.out.println("x <= y = " + (x <= y));
14        System.out.println("x == y = " + (x == y));
15        System.out.println("x != y = " + (x != y));
16
17    }
18 }
19
20 class Relational {
21     public static void main(String[] args) {
22
23         RelationalOperators r = new RelationalOperators();
24         r.operators();
25     }
26 }
```

Output:

```
x = 10
y = 5
x > y = true
x < y = false
x >= y = true
x <= y = false
x == y = false
x != y = true
```

JAVA PROGRAMMING LAB

Experiment No.: 7

Date: 30/07/2026

Aim:

WAP to implement the assignment operators.

Program:

```
1 package Program;
2
3 class AssignmentOperators {
4     void operators() {
5
6         int x = 10, y = 5;
7
8         System.out.println("x = " + x);
9         System.out.println("y = " + y);
10
11         x += y;
12         System.out.println("x += y = " + x);
13
14         x -= y;
15         System.out.println("x -= y = " + x);
16
17         x *= y;
18         System.out.println("x *= y = " + x);
19
20         x /= y;
21         System.out.println("x /= y = " + x);
22
23         x %= y;
24         System.out.println("x %= y = " + x);
25     }
26 }
27
28 class Assignment {
29     public static void main(String[] args) {
30         AssignmentOperators a = new AssignmentOperators();
31         a.operators();
32     }
33 }
```

Output:

```
x = 10
```

```
y = 5  
x += y = 15  
x -= y = 10  
x *= y = 50  
x /= y = 10  
x %= y = 0
```

JAVA PROGRAMMING LAB

Experiment No.: 8

Date: 30/07/2026

Aim:

WAP to implement the increment and decrement operators.

Program:

```
1 package Program;
2
3 class IncrementDecrementOperators {
4     void operators() {
5
6         int x = 10, y = 5;
7
8         System.out.println("x = " + x);
9         System.out.println("y = " + y);
10
11         System.out.println("x++ = " + (x++));
12         System.out.println("x-- = " + (x--));
13         System.out.println("++y = " + (++y));
14         System.out.println("--y = " + (--y));
15     }
16 }
17
18 class IncrementDecrement {
19     public static void main(String[] args) {
20         IncrementDecrementOperators i = new IncrementDecrementOperators();
21         i.operators();
22     }
23 }
```

Output:

```
x = 10
y = 5
x++ = 10
x-- = 11
++y = 6
--y = 5
```

JAVA PROGRAMMING LAB

Experiment No.: 9

Date: 30/07/2026

Aim:

WAP to implement the logical operators.

Program:

```
1 package Program;
2
3 class LogicalOperators {
4     void operator(String[] args) {
5
6         boolean a = true;
7         boolean b = false;
8
9         System.out.println("a = " + a);
10        System.out.println("b = " + b);
11
12        System.out.println("a && b = " + (a && b));
13        System.out.println("a || b = " + (a || b));
14        System.out.println("!a = " + (!a));
15    }
16 }
17
18 class Logical {
19     public static void main(String[] args) {
20
21         LogicalOperators l = new LogicalOperators();
22         l.operator(args);
23     }
24 }
```

Output:

```
a = true
b = false
a && b = false
a || b = true
!a = false
```

JAVA PROGRAMMING LAB

Experiment No.: 10

Date: 30/07/2026

Aim:

WAP to implement the conditional operator.

Program:

```
1 package Program;
2
3 class ConditionalOperators {
4     void operators() {
5         int x = 10, y = 5;
6
7         System.out.println("X = " + x);
8         System.out.println("y = " + y);
9
10        System.out.println("!(x > y) = " + !(x > y));
11        System.out.println("!(x < y) = " + !(x < y));
12        System.out.println("x > y && x < y = " + (x > y && x < y));
13        System.out.println("x > y || x < y = " + (x > y || x < y));
14        System.out.println("x > y ? x : y = " + (x > y ? x : y));
15        System.out.println("x > y ^ x < y = " + (x > y ^ x < y));
16    }
17 }
18
19 class Conditional {
20     public static void main(String[] args) {
21         ConditionalOperators c = new ConditionalOperators();
22         c.operators();
23     }
24 }
```

Output:

```
X = 10
y = 5
!(x > y) = false
!(x < y) = true
x > y && x < y = false
x > y || x < y = true
x > y ? x : y = 10
x > y ^ x < y = true
```

JAVA PROGRAMMING LAB

Experiment No.: 11

Date: 30/07/2026

Aim:

WAP to find the subtraction of two numbers.

Program:

```
1 package Program;
2
3 class Subtraction {
4     void display() {
5         int a = 10, b = 5, c;
6         c = a - b;
7         System.out.println("The subtraction of two numbers: " + c);
8     }
9 }
10
11 }
12
13 class SubtractionDemo {
14     public static void main(String[] args) {
15         Subtraction s = new Subtraction();
16         s.display();
17     }
18 }
19
20 }
```

Output:

```
The subtraction of two numbers: 5
```

JAVA PROGRAMMING LAB

Experiment No.: 12

Date: 30/07/2026

Aim:

WAP to find the multiplication of two numbers.

Program:

```
1 package Program;
2
3 class Multiplication {
4
5     void display() {
6
7         int a = 10, b = 5, c;
8         c = a * b;
9         System.out.println("The multiplication of two numbers: " + c);
10    }
11 }
12
13 class MultiplicationDemo {
14
15     public static void main(String[] args) {
16
17         Multiplication m = new Multiplication();
18         m.display();
19     }
20 }
```

Output:

```
The multiplication of two numbers: 50
```

JAVA PROGRAMMING LAB

Experiment No.: 13

Date: 30/07/2026

Aim:

WAP to find the quotient and remainder of two integers.

Program:

```
1 package Program;
2
3 class Division {
4     void display() {
5         int a = 13, b = 5;
6         System.out.println("The Quotient of two numbers: " + (a / b));
7         System.out.println("The Remainder of two numbers: " + (a % b));
8     }
9 }
10
11 }
12
13 class DivisionDemo {
14     public static void main(String[] args) {
15         Division d = new Division();
16         d.display();
17     }
18 }
19
20 }
```

Output:

```
The Quotient of two numbers: 2
The Remainder of two numbers: 3
```

JAVA PROGRAMMING LAB

Experiment No.: 14

Date: 30/07/2026

Aim:

WAP to calculate the area and perimeter of a rectangle.

Program:

```
1 package Program;
2
3 class Rectangle {
4     void display() {
5         int l = 17, b = 12;
6         System.out.println("The area of rectangle: " + (l * b));
7         System.out.println("The perimeter of rectangle: " + (2 * (l + b)));
8     }
9 }
10
11
12 class RectangleDemo {
13     public static void main(String[] args) {
14         Rectangle r = new Rectangle();
15         r.display();
16     }
17 }
18
19
20 }
```

Output:

```
The area of rectangle: 204
The perimeter of rectangle: 58
```

JAVA PROGRAMMING LAB

Experiment No.: 15

Date: 30/07/2026

Aim:

WAP to calculate the area of a triangle using base and height.

Program:

```
1 package Program;
2
3 class Triangle {
4     void display() {
5         int b = 7, h = 5;
6         System.out.println("The area of triangle: " + (0.5 * b * h));
7     }
8 }
9
10 class TriangleDemo {
11     public static void main(String[] args) {
12         Triangle t = new Triangle();
13         t.display();
14     }
15 }
16
17
18
19 }
```

Output:

```
The area of triangle: 17.5
```

JAVA PROGRAMMING LAB

Experiment No.: 16

Date: 30/07/2026

Aim:

WAP to convert temperature from Celsius to Fahrenheit.

Program:

```
1 package Program;
2
3 class CelciusToFahrnehit {
4
5     void display() {
6
7         int c = 42;
8         System.out.println("The temperature in Celcius: " + c);
9         System.out.println("The temperature in Fahrnehit: " + ((c * 9 / 5) +
10        32));
11     }
12 }
13
14 class CelciusToFahrnehitDemo {
15
16     public static void main(String[] args) {
17
18         CelciusToFahrnehit f = new CelciusToFahrnehit();
19         f.display();
20     }
21 }
```

Output:

```
The temperature in Celcius: 42
The temperature in Fahrnehit: 107
```

JAVA PROGRAMMING LAB

Experiment No.: 17

Date: 30/07/2026

Aim:

WAP to swap two numbers using a third variable.

Program:

```
1 package Program;
2
3 class Swap {
4
5     void display() {
6
7         int a = 11, b = 4, c;
8         System.out.println("The value of a before swapping: " + a);
9         System.out.println("The value of b before swapping: " + b);
10        c = a;
11        a = b;
12        b = c;
13        System.out.println("\nThe value of a after swapping: " + a);
14        System.out.println("The value of b after swapping: " + b);
15    }
16 }
17
18 class SwapDemo {
19
20     public static void main(String[] args) {
21
22         Swap s = new Swap();
23         s.display();
24     }
25 }
```

Output:

```
The value of a before swapping: 11
The value of b before swapping: 4

The value of a after swapping: 4
The value of b after swapping: 11
```

JAVA PROGRAMMING LAB

Experiment No.: 18

Date: 30/07/2026

Aim:

WAP to swap two numbers without using a third variable.

Program:

```
1 package Program;
2
3 class SwapWithoutTemp {
4
5     void display() {
6
7         int a = 17, b = 6;
8         System.out.println("The value of a before swapping: " + a);
9         System.out.println("The value of b before swapping: " + b);
10        a = a + b;
11        b = a - b;
12        a = a - b;
13        System.out.println("\nThe value of a after swapping: " + a);
14        System.out.println("The value of b after swapping: " + b);
15    }
16 }
17
18 class SwapWithoutTempDemo {
19
20     public static void main(String[] args) {
21
22         SwapWithoutTemp s = new SwapWithoutTemp();
23         s.display();
24     }
25 }
```

Output:

```
The value of a before swapping: 17
The value of b before swapping: 6

The value of a after swapping: 6
The value of b after swapping: 17
```

JAVA PROGRAMMING LAB

Experiment No.: 19

Date: 30/07/2026

Aim:

WAP to calculate simple interest using principal, rate and time.

Program:

```
1 package Program;
2
3 class SimpleInterest {
4
5     void display() {
6
7         int p = 10000, r = 7;
8         float t = 2.5f, si;
9         si = (p * r * t) / 100;
10        System.out.println("The simple interest is: " + si);
11    }
12 }
13
14 class SimpleInterestDemo {
15
16     public static void main(String[] args) {
17
18         SimpleInterest s = new SimpleInterest();
19         s.display();
20     }
21 }
```

Output:

```
The simple interest is: 1750.0
```

JAVA PROGRAMMING LAB

Experiment No.: 20

Date: 30/07/2026

Aim:

WAP to find the greater of two numbers using the conditional (?:) operator.

Program:

```
1 package Program;
2
3 class GreaterNumebr {
4
5     void display() {
6
7         int a = 23, b = 45;
8         System.out.println("The greater number: " + (a > b ? a : b));
9     }
10 }
11
12 class GreaterNumebrDemo {
13
14     public static void main(String[] args) {
15
16         GreaterNumebr g = new GreaterNumebr();
17         g.display();
18     }
19 }
```

Output:

```
The greater number: 45
```

JAVA PROGRAMMING LAB

Experiment No.: 21

Date: 06/08/2026

Aim:

WAP to implement the following Math functions:

- sqrt(x)
- pow(x, y)
- sin(x)
- cos(x)
- tan(x)
- ceil(x)
- floor(x)
- rint(x)
- abs(a)
- max(a, b)
- min(a, b)

Program:

```
1 package Program;
2
3 class MathFunctions {
4
5     void math() {
6
7         double x = 25.5;
8         double y = 2;
9         int a = -10;
10        int b = 20;
11
12        System.out.println("sqrt(x) = " + Math.sqrt(x));
13        System.out.println("pow(x, y) = " + Math.pow(x, y));
14
15        System.out.println("sin(x) = " + Math.sin(x));
16        System.out.println("cos(x) = " + Math.cos(x));
17        System.out.println("tan(x) = " + Math.tan(x));
18    }
```

```

19         System.out.println("ceil(x) = " + Math.ceil(x));
20         System.out.println("floor(x) = " + Math.floor(x));
21         System.out.println("rint(x) = " + Math.rint(x));
22
23         System.out.println("abs(a) = " + Math.abs(a));
24
25         System.out.println("max(a, b) = " + Math.max(a, b));
26         System.out.println("min(a, b) = " + Math.min(a, b));
27     }
28 }
29
30 class MathDemo {
31
32     public static void main(String[] args) {
33
34         MathFunctions m = new MathFunctions();
35         m.math();
36     }
37 }

```

Output:

```

sqrt(x) = 5.049752469181039
pow(x, y) = 650.25
sin(x) = 0.3590583540221683
cos(x) = 0.933315112063922
tan(x) = 0.3847128899779099
ceil(x) = 26.0
floor(x) = 25.0
rint(x) = 26.0
abs(a) = 10
max(a, b) = 20
min(a, b) = -10

```

JAVA PROGRAMMING LAB

Experiment No.: 22

Date: 06/08/2026

Aim:

WAP to implement only if statement.

Program:

```
1 package Program;
2
3 class CheckNumber {
4     void check() {
5         int number = 10;
6
7         if (number > 0) {
8             System.out.println("The number is positive.");
9         }
10    }
11 }
12
13
14
15
16 class IfStatement {
17     public static void main(String[] args) {
18         CheckNumber c = new CheckNumber();
19         c.check();
20     }
21 }
22
23 }
```

Output:

```
The number is positive.
```

JAVA PROGRAMMING LAB

Experiment No.: 23

Date: 06/08/2026

Aim:

WAP to implement if-else statement.

Program:

```
1 package Program;
2
3 class CheckNumbers {
4
5     void check() {
6
7         int number = 15;
8
9         if (number % 2 == 0) {
10             System.out.println(number + " is an even number.");
11         } else {
12             System.out.println(number + " is an odd number.");
13         }
14     }
15 }
16
17 class EvenOdd {
18
19     public static void main(String[] args) {
20
21         CheckNumbers c = new CheckNumbers();
22         c.check();
23     }
24 }
```

Output:

```
15 is an odd number.
```

JAVA PROGRAMMING LAB

Experiment No.: 24

Date: 06/08/2026

Aim:

WAP to implement nested if-else statement.

Program:

```
1 package Program;
2
3 class GreatestNumber {
4
5     void check() {
6
7         int a = 10;
8         int b = 25;
9         int c = 15;
10
11         if (a > b) {
12
13             if (a > c) {
14                 System.out.println(a + " is the greatest number.");
15             } else {
16                 System.out.println(c + " is the greatest number.");
17             }
18
19         } else {
20
21             if (b > c) {
22                 System.out.println(b + " is the greatest number.");
23             } else {
24                 System.out.println(c + " is the greatest number.");
25             }
26         }
27     }
28 }
29
30 class NestedIfElse {
31
32     public static void main(String[] args) {
33
34         GreatestNumber g = new GreatestNumber();
35         g.check();
36     }
37 }
```

Output:

```
25 is the greatest number.
```

JAVA PROGRAMMING LAB

Experiment No.: 25

Date: 06/08/2026

Aim:

WAP to implement if else ladder.

Program:

```
1 package Program;
2
3 class Grade {
4
5     void check() {
6
7         int marks = 75;
8
9         if (marks >= 90) {
10             System.out.println("Grade A+");
11         } else if (marks >= 80) {
12             System.out.println("Grade A");
13         } else if (marks >= 70) {
14             System.out.println("Grade B");
15         } else if (marks >= 60) {
16             System.out.println("Grade C");
17         } else if (marks >= 50) {
18             System.out.println("Grade D");
19         } else if (marks >= 40) {
20             System.out.println("Grade E");
21         } else {
22             System.out.println("Grade F");
23         }
24     }
25 }
26
27 class ElseIfLadder {
28
29     public static void main(String[] args) {
30
31         Grade g = new Grade();
32         g.check();
33     }
34 }
```

Output:

Grade B

JAVA PROGRAMMING LAB

Experiment No.: 26

Date: 06/08/2026

Aim:

WAP to test the given number is positive or negative.

Program:

```
1 package Program;
2
3 class CheckNumber {
4
5     void check() {
6
7         int number = -25;
8
9         if (number > 0) {
10             System.out.println(number + " is Positive.");
11         } else if (number < 0) {
12             System.out.println(number + " is Negative.");
13         } else {
14             System.out.println("The number is Zero.");
15         }
16     }
17 }
18
19 class PositiveNegative {
20
21     public static void main(String[] args) {
22
23         CheckNumber c = new CheckNumber();
24         c.check();
25     }
26 }
```

Output:

The number is positive.

JAVA PROGRAMMING LAB

Experiment No.: 27

Date: 06/08/2026

Aim:

WAP to implement if-else statement.

Program:

```
1 package Program;
2
3 class CheckAge {
4     void check() {
5         int age = 20;
6
7         if (age >= 18) {
8             System.out.println("You are eligible to vote.");
9         } else {
10             System.out.println("You are not eligible to vote.");
11         }
12     }
13 }
14
15 }
16
17 class IfElse {
18     public static void main(String[] args) {
19         CheckAge a = new CheckAge();
20         a.check();
21     }
22 }
23
24 }
```

Output:

```
You are eligible to vote.
```

JAVA PROGRAMMING LAB

Experiment No.: 28

Date: 06/08/2026

Aim:

WAP to display numbers from 1 to 10 using do-while loop.

Program:

```
1 package Program;
2
3 class Display {
4
5     void numbers() {
6
7         int i = 1;
8
9         do {
10             System.out.println(i);
11             i++;
12         } while (i <= 10);
13     }
14 }
15
16 class DoWhile {
17
18     public static void main(String[] args) {
19
20         Display d = new Display();
21         d.numbers();
22     }
23 }
```

Output:

```
1
2
3
4
5
6
7
8
9
10
```

JAVA PROGRAMMING LAB

Experiment No.: 29

Date: 06/08/2026

Aim:

WAP to display numbers from 1 to 10 using while loop.

Program:

```
1 package Program;
2
3 class Display {
4
5     void numbers() {
6
7         int i = 1;
8
9         while (i <= 10) {
10             System.out.println(i);
11             i++;
12         }
13     }
14 }
15
16 class WhileLoop {
17
18     public static void main(String[] args) {
19
20         Display d = new Display();
21         d.numbers();
22     }
23 }
```

Output:

```
1
2
3
4
5
6
7
8
9
10
```

JAVA PROGRAMMING LAB

Experiment No.: 30

Date: 06/08/2026

Aim:

WAP to display numbers from 1 to 10 using for loop.

Program:

```
1 package Program;
2
3 class Display {
4     void numbers() {
5         for (int i = 1; i <= 10; i++) {
6             System.out.println(i);
7         }
8     }
9 }
10
11 class ForLoop {
12     public static void main(String[] args) {
13         Display d = new Display();
14         d.numbers();
15     }
16 }
17
18
19
20
```

Output:

```
1
2
3
4
5
6
7
8
9
10
```

JAVA PROGRAMMING LAB

Experiment No.: 31

Date: 06/08/2026

Aim:

WAP to find the greater of three numbers using if-else statements.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class GreaterNumber {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter three numbers: ");
12         int a = sc.nextInt();
13         int b = sc.nextInt();
14         int c = sc.nextInt();
15
16         sc.close();
17
18         if (a >= b && a >= c) {
19             System.out.println("The greatest number is: " + a);
20         } else if (b >= a && b >= c) {
21             System.out.println("The greatest number is: " + b);
22         } else {
23             System.out.println("The greatest number is: " + c);
24         }
25     }
26 }
27
28 class GreaterNumberDemo {
29
30     public static void main(String[] args) {
31
32         GreaterNumber g = new GreaterNumber();
33         g.display();
34     }
35 }
```

Output:

```
Enter three numbers: 7 3 9
The greatest number is: 9
```

JAVA PROGRAMMING LAB

Experiment No.: 32

Date: 06/08/2026

Aim:

WAP to check whether a given number is divisible by 5 and 11 using if-else statement.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Divisible {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a number: ");
12         int a = sc.nextInt();
13
14         sc.close();
15
16         if (a % 7 == 0 && a % 11 == 0) {
17             System.out.println("The number is divisible by 7 and 11.");
18         } else {
19             System.out.println("The number is not divisible by 7 and 11.");
20         }
21     }
22 }
23
24 class DivisibleDemo {
25
26     public static void main(String[] args) {
27
28         Divisible d = new Divisible();
29         d.display();
30     }
31 }
```

Output:

```
Enter a number: 154
The number is divisible by 7 and 11.
```

JAVA PROGRAMMING LAB

Experiment No.: 33

Date: 06/08/2026

Aim:

WAP to check whether an entered character is a vowel or consonant using if-else statement.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class VowelOrConsonant {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a character: ");
12         char ch = sc.next().charAt(0);
13
14         sc.close();
15
16         if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
17         ch == 'A' || ch == 'E' || ch == 'I'
18             || ch == 'O' || ch == 'U') {
19             System.out.println("The character is a vowel.");
20         } else {
21             System.out.println("The character is a consonant.");
22         }
23     }
24 }
25
26 class VowelOrConsonantDemo {
27
28     public static void main(String[] args) {
29
30         VowelOrConsonant v = new VowelOrConsonant();
31         v.display();
32     }
33 }
```

Output:

```
Enter a character: t
The character is a consonant.
```

JAVA PROGRAMMING LAB

Experiment No.: 34

Date: 06/08/2026

Aim:

WAP to display the multiplication table of a given number using for loop.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class MultiplicationTable {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a number to print its multiplication table: ")
12         ;
13         int num = sc.nextInt();
14
15         System.out.print("Enter the range of multiplication table: ");
16         int r = sc.nextInt();
17
18         sc.close();
19
20         for (int i = 0; i <= r; i++) {
21             System.out.println(" " + num + " * " + i + " = " + (num * i));
22         }
23     }
24
25 class MultiplicationTableDemo {
26
27     public static void main(String[] args) {
28
29         MultiplicationTable m = new MultiplicationTable();
30         m.display();
31     }
32 }
```

Output:

```
Enter a number to print its multiplication table: 13
```

Enter the range of multiplication table: 10

$$13 * 0 = 0$$

$$13 * 1 = 13$$

$$13 * 2 = 26$$

$$13 * 3 = 39$$

$$13 * 4 = 52$$

$$13 * 5 = 65$$

$$13 * 6 = 78$$

$$13 * 7 = 91$$

$$13 * 8 = 104$$

$$13 * 9 = 117$$

$$13 * 10 = 130$$

JAVA PROGRAMMING LAB

Experiment No.: 35

Date: 06/08/2026

Aim:

WAP to find the sum of numbers from 1 to 10 using for loop.

Program:

```
1 package Program;
2
3 class Sum1To10 {
4
5     void display() {
6
7         int sum = 0;
8
9         for (int i = 1; i <= 10; i++) {
10             sum += i;
11         }
12         System.out.println("The sum of numbers from 1 to 10 is: " + sum);
13     }
14 }
15
16 class Sum1To10Demo {
17
18     public static void main(String[] args) {
19
20         Sum1To10 s = new Sum1To10();
21         s.display();
22     }
23 }
```

Output:

```
The sum of numbers from 1 to 10 is: 55
```

JAVA PROGRAMMING LAB

Experiment No.: 36

Date: 06/08/2026

Aim:

WAP to display all even numbers from 1 to 10 using while loop.

Program:

```
1 package Program;
2
3 class Even1To10 {
4     void display() {
5         System.out.print("Even numbers from 1 to 10 are: ");
6
7         int i = 1;
8         while (i <= 10) {
9             if (i % 2 == 0) {
10                System.out.print(i + " ");
11            }
12            i++;
13        }
14        System.out.println();
15    }
16 }
17
18 }
19
20
21 class Even1To10Demo {
22     public static void main(String[] args) {
23
24         Even1To10 e = new Even1To10();
25         e.display();
26     }
27 }
28 }
```

Output:

```
Even numbers from 1 to 10 are: 2 4 6 8 10
```

JAVA PROGRAMMING LAB

Experiment No.: 37

Date: 06/08/2026

Aim:

WAP to display all odd numbers from 1 to 10 using do-while loop.

Program:

```
1 package Program;
2
3 class Odd1To10 {
4
5     void display() {
6
7         System.out.print("Odd numbers from 1 to 10 are: ");
8
9         int i = 1;
10        do {
11            if (i % 2 != 0) {
12                System.out.print(i + " ");
13            }
14            i++;
15        } while (i <= 10);
16        System.out.println();
17    }
18 }
19
20 class Odd1To10Demo {
21
22     public static void main(String[] args) {
23
24         Odd1To10 o = new Odd1To10();
25         o.display();
26     }
27 }
```

Output:

```
Odd numbers from 1 to 10 are: 1 3 5 7 9
```

JAVA PROGRAMMING LAB

Experiment No.: 38

Date: 06/08/2026

Aim:

WAP to calculate the sum of even and odd numbers separately from 1 to 100.

Program:

```
1 package Program;
2
3 class SumEven1To100 {
4
5     void display() {
6
7         int even = 0;
8         for (int i = 1; i <= 100; i++) {
9             if (i % 2 == 0) {
10                 even += i;
11             }
12         }
13         System.out.print("Sum of even number between 1 to 100: " + even + "\n"
14     );
15     }
16 }
17
18 class SumOdd1To100 {
19
20     void display() {
21
22         int odd = 0;
23         for (int i = 1; i <= 100; i++) {
24             if (i % 2 != 0) {
25                 odd += i;
26             }
27         }
28         System.out.println("Sum of odd number between 1 to 100: " + odd);
29         System.out.println();
30     }
31 }
32
33 class SumDemo {
34
35     public static void main(String[] args) {
36
37         SumEven1To100 e = new SumEven1To100();
38         e.display();
39     }
40 }
```

```
38
39         SumOdd1To100 o = new SumOdd1To100();
40         o.display();
41     }
42 }
```

Output:

```
Sum of even number between 1 to 100: 2550
Sum of odd number between 1 to 100: 2500
```

JAVA PROGRAMMING LAB

Experiment No.: 39

Date: 06/08/2026

Aim:

WAP to find the sum of digits of a given number using while loop.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class SumOfDigit {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a number of multiple digits: ");
12         int num = sc.nextInt();
13
14         sc.close();
15
16         int demo = num;
17         int sum = 0;
18
19         while (num != 0) {
20             sum += (num % 10);
21             num /= 10;
22         }
23
24         System.out.print("Sum of digits of number " + demo + ": " + sum + "\n"
25     );
26 }
27
28 class SumOfDigitDemo {
29
30     public static void main(String[] args) {
31
32         SumOfDigit s = new SumOfDigit();
33         s.display();
34     }
35 }
```

Output:

Enter a number of multiple digits: 154
Sum of digits of number 154: 10

JAVA PROGRAMMING LAB

Experiment No.: 40

Date: 06/08/2026

Aim:

WAP to reverse a given integer using while loop.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class Reverse {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a number to reverse: ");
12         int num = sc.nextInt();
13
14         sc.close();
15
16         int demo = num;
17         int rev = 0;
18
19         while (num != 0) {
20             rev = rev * 10 + num % 10;
21             num /= 10;
22         }
23
24         System.out.print("Reverse of number " + demo + ": " + rev + "\n");
25     }
26 }
27
28 class ReverseDemo {
29
30     public static void main(String[] args) {
31         Reverse r = new Reverse();
32         r.display();
33     }
34 }
```

Output:

```
Enter a number to reverse: 45  
Reverse of number 45: 54
```

JAVA PROGRAMMING LAB

Experiment No.: 41

Date: 13/08/2026

Aim:

WAP to display stars in a triangular form a single star in 1st line, two stars in 2nd line, three stars in 3rd line and so on.

Program:

```
1 package Program;
2
3 class StarTriangle {
4
5     void display() {
6
7         for (int i = 1; i <= 5; i++) {
8
9             for (int j = 1; j <= i; j++) {
10                 System.out.print("* ");
11             }
12
13             System.out.println();
14         }
15     }
16 }
17
18 class StarTriangleDemo {
19
20     public static void main(String[] args) {
21
22         StarTriangle s = new StarTriangle();
23         s.display();
24     }
25 }
```

Output:

```
*
* *
* * *
* * * *
* * * * *
```

JAVA PROGRAMMING LAB

Experiment No.: 42

Date: 13/08/2026

Aim:

WAP to implement switch statement.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class SwitchStatement {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a number (1-3): ");
12         int choice = sc.nextInt();
13
14         switch (choice) {
15
16             case 1:
17                 System.out.println("You selected One.");
18                 break;
19
20             case 2:
21                 System.out.println("You selected Two.");
22                 break;
23
24             case 3:
25                 System.out.println("You selected Three.");
26                 break;
27
28             default:
29                 System.out.println("Invalid choice.");
30         }
31
32         sc.close();
33     }
34 }
35
36 class SwitchStatementDemo {
37
38     public static void main(String[] args) {
```

```
39
40     SwitchStatement s = new SwitchStatement();
41     s.display();
42 }
43 }
```

Output:

```
Enter a number (1-3): 3
You selected Three.
```

JAVA PROGRAMMING LAB

Experiment No.: 43

Date: 13/08/2026

Aim:

WAP to display numbers from 10 to 1 in descending order using for loop.

Program:

```
1 package Program;
2
3 class DescendingNumbers {
4
5     void display() {
6
7         for (int i = 10; i >= 1; i--) {
8             System.out.println(i);
9         }
10    }
11 }
12
13 class DescendingNumbersDemo {
14
15     public static void main(String[] args) {
16
17         DescendingNumbers d = new DescendingNumbers();
18         d.display();
19     }
20 }
```

Output:

```
10
9
8
7
6
5
4
3
2
1
```

JAVA PROGRAMMING LAB

Experiment No.: 44

Date: 13/08/2026

Aim:

WAP to display numbers from 1 to 10 using infinite for loop.

Program:

```
1 package Program;
2
3 class InfiniteFor {
4     void display() {
5         for (int i = 1;; i++) {
6
7             System.out.println(i);
8
9             if (i == 10) {
10                break;
11            }
12        }
13    }
14 }
15
16
17 class InfiniteForDemo {
18     public static void main(String[] args) {
19
20         InfiniteFor i = new InfiniteFor();
21         i.display();
22     }
23 }
24
25 }
```

Output:

```
1
2
3
4
5
6
7
8
9
```


JAVA PROGRAMMING LAB

Experiment No.: 45

Date: 13/08/2026

Aim:

WAP to accept and display a character from the keyboard using Buffered Reader class.

Program:

```
1 package Program;
2
3 import java.io.BufferedReader;
4 import java.io.IOException;
5 import java.io.InputStreamReader;
6
7 class CharacterInput {
8
9     void display() throws IOException {
10
11         BufferedReader br = new BufferedReader(
12             new InputStreamReader(System.in));
13
14         System.out.print("Enter a character: ");
15         char ch = br.readLine().charAt(0);
16
17         System.out.println("The entered character is: " + ch);
18     }
19 }
20
21 class CharacterInputDemo {
22
23     public static void main(String[] args) throws IOException {
24
25         CharacterInput c = new CharacterInput();
26         c.display();
27     }
28 }
```

Output:

```
Enter a character: d
The entered character is: d
```

JAVA PROGRAMMING LAB

Experiment No.: 46

Date: 13/08/2026

Aim:

WAP to accept and display a string or a name from the keyboard using Buffered Reader class.

Program:

```
1 package Program;
2
3 import java.io.*;
4
5 class StringInput {
6
7     void display() throws IOException {
8
9         BufferedReader br = new BufferedReader(
10             new InputStreamReader(System.in));
11
12         System.out.print("Enter your name: ");
13         String name = br.readLine();
14
15         System.out.println("Your name is: " + name);
16     }
17 }
18
19 class StringInputDemo {
20
21     public static void main(String[] args) throws IOException {
22
23         StringInput s = new StringInput();
24         s.display();
25     }
26 }
```

Output:

```
Enter your name: Aditya Raj
Your name is: Aditya Raj
```

JAVA PROGRAMMING LAB

Experiment No.: 47

Date: 13/08/2026

Aim:

WAP to accept and display an integer from the keyboard using Buffered Reader class.

Program:

```
1 package Program;
2
3 import java.io.*;
4
5 class IntegerInput {
6
7     void display() throws IOException {
8
9         BufferedReader br = new BufferedReader(
10             new InputStreamReader(System.in));
11
12         System.out.print("Enter an integer: ");
13         int num = Integer.parseInt(br.readLine());
14
15         System.out.println("The entered integer is: " + num);
16     }
17 }
18
19 class IntegerInputDemo {
20
21     public static void main(String[] args) throws IOException {
22
23         IntegerInput i = new IntegerInput();
24         i.display();
25     }
26 }
```

Output:

```
Enter an integer: 23
The entered integer is: 23
```

JAVA PROGRAMMING LAB

Experiment No.: 48

Date: 13/08/2026

Aim:

WAP to accept and display a float number from the keyboard using Buffered Reader class.

Program:

```
1 package Program;
2
3 import java.io.*;
4
5 class FloatInput {
6
7     void display() throws IOException {
8
9         BufferedReader br = new BufferedReader(
10             new InputStreamReader(System.in));
11
12         System.out.print("Enter a float number: ");
13         float num = Float.parseFloat(br.readLine());
14
15         System.out.println("The entered float number is: " + num);
16     }
17 }
18
19 class FloatInputDemo {
20
21     public static void main(String[] args) throws IOException {
22
23         FloatInput f = new FloatInput();
24         f.display();
25     }
26 }
```

Output:

```
Enter a float number: 43.67
The entered float number is: 43.67
```

JAVA PROGRAMMING LAB

Experiment No.: 49

Date: 13/08/2026

Aim:

WAP to accept and display employee details using Buffered Reader class.

Program:

```
1 package Program;
2
3 import java.io.*;
4
5 class EmployeeDetails {
6
7     void display() throws IOException {
8
9         BufferedReader br = new BufferedReader(
10             new InputStreamReader(System.in));
11
12         System.out.print("Enter employee ID: ");
13         int id = Integer.parseInt(br.readLine());
14
15         System.out.print("Enter employee name: ");
16         String name = br.readLine();
17
18         System.out.print("Enter employee salary: ");
19         float salary = Float.parseFloat(br.readLine());
20
21         System.out.println("\nEmployee Details");
22         System.out.println("Employee ID: " + id);
23         System.out.println("Employee Name: " + name);
24         System.out.println("Employee Salary: " + salary);
25     }
26 }
27
28 class EmployeeDetailsDemo {
29
30     public static void main(String[] args) throws IOException {
31
32         EmployeeDetails e = new EmployeeDetails();
33         e.display();
34     }
35 }
```

Output:

```
Enter employee ID: 101
Enter employee name: Ravi
Enter employee salary: 75000
```

```
Employee Details
Employee ID: 101
Employee Name: Ravi
Employee Salary: 75000.0
```

JAVA PROGRAMMING LAB

Experiment No.: 50

Date: 13/08/2026

Aim:

WAP accept a year from the keyboard and test if it is leap or not.

Program:

```
1 package Program;
2
3 import java.util.Scanner;
4
5 class LeapYear {
6
7     void display() {
8
9         Scanner sc = new Scanner(System.in);
10
11         System.out.print("Enter a year: ");
12         int year = sc.nextInt();
13
14         sc.close();
15
16         if (year % 400 == 0) {
17             System.out.println(year + " is a leap year.");
18         } else if (year % 100 == 0) {
19             System.out.println(year + " is not a leap year.");
20         } else if (year % 4 == 0) {
21             System.out.println(year + " is a leap year.");
22         } else {
23             System.out.println(year + " is not a leap year.");
24         }
25     }
26 }
27
28 class LeapYearDemo {
29
30     public static void main(String[] args) {
31
32         LeapYear l = new LeapYear();
33         l.display();
34     }
35 }
```

Output:

```
Enter a year: 2020
2020 is a leap year.
```

JAVA PROGRAMMING LAB

Experiment No.: 51

Date: 13/08/2026

Aim:

WAP to display the following number pattern using nested for loops. 1 1 2 1 2 3 1 2
3 4 1 2 3 4 5

Program:

```
1 package Program;
2
3 class Pattern1 {
4
5     void display() {
6
7         int i, j;
8
9         for (i = 1; i <= 5; i++) {
10             for (j = 1; j <= i; j++) {
11                 System.out.print(j);
12             }
13             System.out.println();
14         }
15     }
16 }
17
18 class P1 {
19
20     public static void main(String[] args) {
21
22         Pattern1 p1 = new Pattern1();
23         p1.display();
24     }
25 }
```

Output:

```
1
12
123
1234
12345
```

JAVA PROGRAMMING LAB

Experiment No.: 52

Date: 13/08/2026

Aim:

WAP to display the following number pattern using nested for loops. 1 2 2 3 3 3 4 4 4 5 5 5 5

Program:

```
1 package Program;
2
3 class Pattern2 {
4
5     void display() {
6
7         int i, j;
8         for (i = 1; i <= 5; i++) {
9             for (j = 1; j <= i; j++) {
10                 System.out.print(i);
11             }
12             System.out.println();
13         }
14     }
15 }
16
17 class P2 {
18
19     public static void main(String[] args) {
20
21         Pattern2 p2 = new Pattern2();
22         p2.display();
23     }
24 }
```

Output:

```
1
22
333
4444
55555
```

JAVA PROGRAMMING LAB

Experiment No.: 53

Date: 13/08/2026

Aim:

WAP to display the following reverse number pattern using nested for loops. 5 5 5 5
5 4 4 4 4 3 3 3 2 2 1

Program:

```
1 package Program;
2
3 class Pattern3 {
4
5     void display() {
6
7         int i, j;
8         for (i = 5; i >= 1; i--) {
9             for (j = 1; j <= i; j++) {
10                 System.out.print(i);
11             }
12             System.out.println();
13         }
14     }
15 }
16
17 class P3 {
18
19     public static void main(String[] args) {
20
21         Pattern3 p3 = new Pattern3();
22         p3.display();
23     }
24 }
```

Output:

```
55555
4444
333
22
1
```

JAVA PROGRAMMING LAB

Experiment No.: 54

Date: 13/08/2026

Aim:

WAP to display the following reverse number pattern using nested for loops. 1 2 3 4
5 1 2 3 4 1 2 3 1 2 1

Program:

```
1 package Program;
2
3 class Patter4 {
4
5     void display() {
6
7         int i, j;
8         for (i = 5; i >= 1; i--) {
9             for (j = 1; j <= i; j++) {
10                 System.out.print(j);
11             }
12             System.out.println();
13         }
14     }
15 }
16
17 class P4 {
18
19     public static void main(String[] args) {
20
21         Patter4 p4 = new Patter4();
22         p4.display();
23     }
24 }
```

Output:

```
12345
1234
123
12
1
```

JAVA PROGRAMMING LAB

Experiment No.: 55

Date: 13/08/2026

Aim:

WAP to display a pyramid pattern of stars using nested for loops.

Program:

```
1 package Program;
2
3 class Pattern5 {
4     void display() {
5
6         int i, j, k;
7         for (i = 1; i <= 15; i++) {
8
9             for (j = 1; j <= 15 - i; j++) {
10                 System.out.print(" ");
11             }
12
13             for (k = 1; k <= 2 * i - 1; k++) {
14                 System.out.print("*");
15             }
16
17             System.out.println();
18         }
19     }
20 }
21
22 class P5 {
23
24     public static void main(String[] args) {
25
26         Pattern5 p5 = new Pattern5();
27         p5.display();
28     }
29 }
```

Output:

```
      *
     ***
    *****
   ********
  *********
 *****
```

```
*****  
*****  
*****  
*****  
*****  
*****  
*****  
*****  
*****  
*****
```